



DS361 – Machine Learning

ClimaPredict Pakistan

Machine Learning Based Weather Prediction System (2023)

Project Report

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I. ABSTRACT

Climate change and unpredictable weather patterns have increased the need for accurate weather forecasting systems. This project, titled "ClimaPredict Pakistan: Machine Learning Based Weather Prediction System", presents an intelligent weather prediction framework developed using historical meteorological data collected from six major cities of Pakistan including Islamabad, Karachi, Lahore, Multan, Peshawar, and Quetta for the year 2023.

The system utilizes Machine Learning techniques to perform three prediction tasks: temperature forecasting through regression, rainfall prediction through binary classification, and weather condition prediction through multi-class classification. A comprehensive dataset containing 2,190 daily records and 38 meteorological features such as temperature, humidity, rainfall, cloud cover, wind speed, air pressure, visibility, and air quality indicators was processed and analyzed.

Exploratory Data Analysis (EDA) was conducted to identify seasonal patterns, rainfall trends, and relationships among variables. Feature engineering techniques including lag features, rolling averages, label encoding, and cyclic encoding were applied to improve predictive performance. Multiple machine learning algorithms including Linear Regression, Logistic Regression, Random Forest, Gradient Boosting, and XGBoost were trained and evaluated.

Experimental results demonstrated that Linear Regression achieved the best performance for temperature prediction, Random Forest produced the highest accuracy for rainfall prediction, and Gradient Boosting performed best for weather condition classification. An interactive web-based dashboard was also developed using HTML, CSS, and JavaScript to allow users to visualize weather trends and generate real-time predictions.

The project demonstrates the effectiveness of machine learning in weather forecasting and provides a scalable framework that can be extended to support real-time weather prediction and nationwide climate analysis.

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1. Project Overview

1.1. Introduction

ClimaPredict Pakistan is a machine learning project that forecasts weather conditions across six major Pakistani cities using historical meteorological data spanning of whole 2023. The dataset was collected using the Open-Meteo Historical Weather API (open-meteo.com), a free and open-source weather API providing high-resolution historical data. The system tackles three distinct prediction tasks — temperature regression, rain classification, and full weather condition prediction — providing a comprehensive multi-task ML pipeline useful for agricultural planning, infrastructure management, and public safety.

1.2. Objectives

- Predict daily average temperature (regression) with high accuracy
- Classify whether a day will receive rainfall (binary classification)
- Predict overall weather condition: Sunny, Rainy, Moderate, or Cloudy (multi-class classification)
- Develop a user-facing interactive dashboard for real-time predictions
- Evaluate and compare multiple ML algorithms across all three tasks

1.3. Scope

The project covers 6 major cities: Islamabad, Karachi, Lahore, Multan, Peshawar, and Quetta, with data collected daily over a one-year period (2023). The dataset contains 2,190 rows and 38 features including temperature, humidity, air pressure, cloud cover, wind speed, visibility, and air quality indices.

2. Dataset Description

The Pakistan Weather Dataset was collected using the Open-Meteo Historical Weather API (open-meteo.com) — a free, open-source meteorological API that provides high-resolution historical weather data. API calls were made for each of the 6 cities across the full 2023 date range, and the responses were compiled into a structured CSV dataset containing 2,190 daily records.

2.1. Dataset Summary

Attribute	Value
Data Source	Open-Meteo Historical Weather API (open-meteo.com)
Total Records	2,190 rows
Total Features	38 columns

Date Range	January 2023 — December 2023
Cities	Islamabad, Karachi, Lahore, Multan, Peshawar, Quetta
Target Variables	AvgTemp, RainCategory, WeatherCondition
API Type	Free & Open-Source REST API — No authentication required
Missing Values	Minimal (< 1% of total cells)

2.2. Key Features

- Temperature: MaxTemp, MinTemp, AvgTemp, FeelsLikeMax, FeelsLikeMin, TempCategory
- Humidity & Rainfall: Humidity, Precipitation, Rainfall, Snowfall, RainCategory
- Atmospheric: AirPressure, DewPoint, CloudCover, Visibility
- Wind: MaxWindSpeed, WindDirection
- Time: Date, Day, Month, Season, Weekend
- Air Quality: PM2_5, PM10, CO_Level
- Lag / Rolling Features: Temp_Yesterday, 7DayAvgTemp, Humidity_Yesterday, 7DayAvgHumidity, Rainfall_Yesterday, 7DayAvgRainfall

3. Exploratory Data Analysis (EDA)

Comprehensive EDA was performed to understand the data distribution, seasonal patterns, inter-feature correlations, and target variable characteristics.

3.1. Temperature Analysis

- Average temperatures peak in May–July (summer), reaching above 35°C in cities like Multan and Karachi
- Quetta and Islamabad record lower average temperatures due to elevation
- Temperature categories: Cold, Mild, Warm, and Hot — distributed roughly evenly
- Clear seasonal patterns observed — Winter (Dec–Feb) vs Summer (May–Jul)

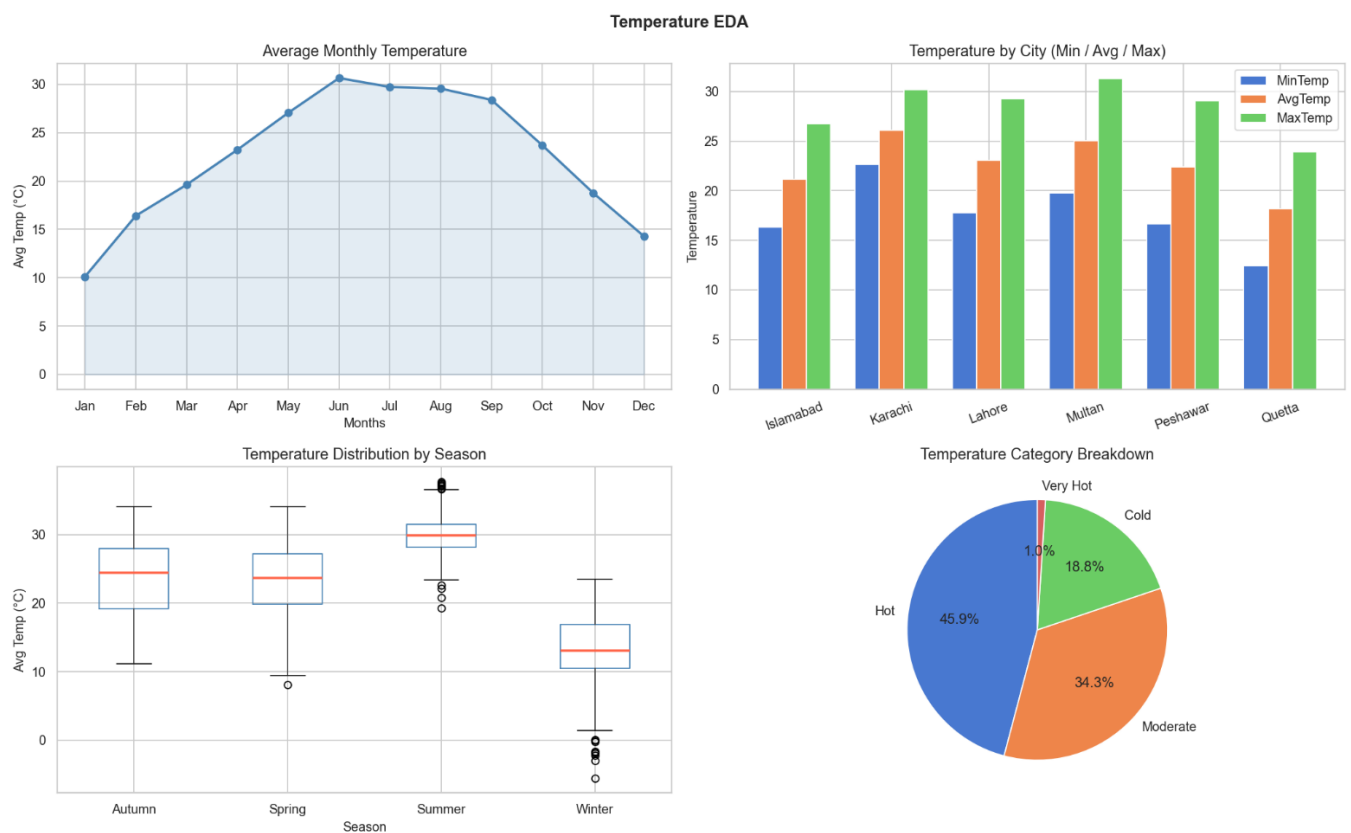


Fig. Temperature Trend Analysis

3.2. Rainfall and Humidity Analysis

- Monsoon season (Summer) accounts for the majority of annual rainfall
- Karachi shows highest average humidity(67.9%) among all six cities
- Strong positive correlation between humidity and rainfall
- Cloud cover is a strong predictor of rainfall occurrence

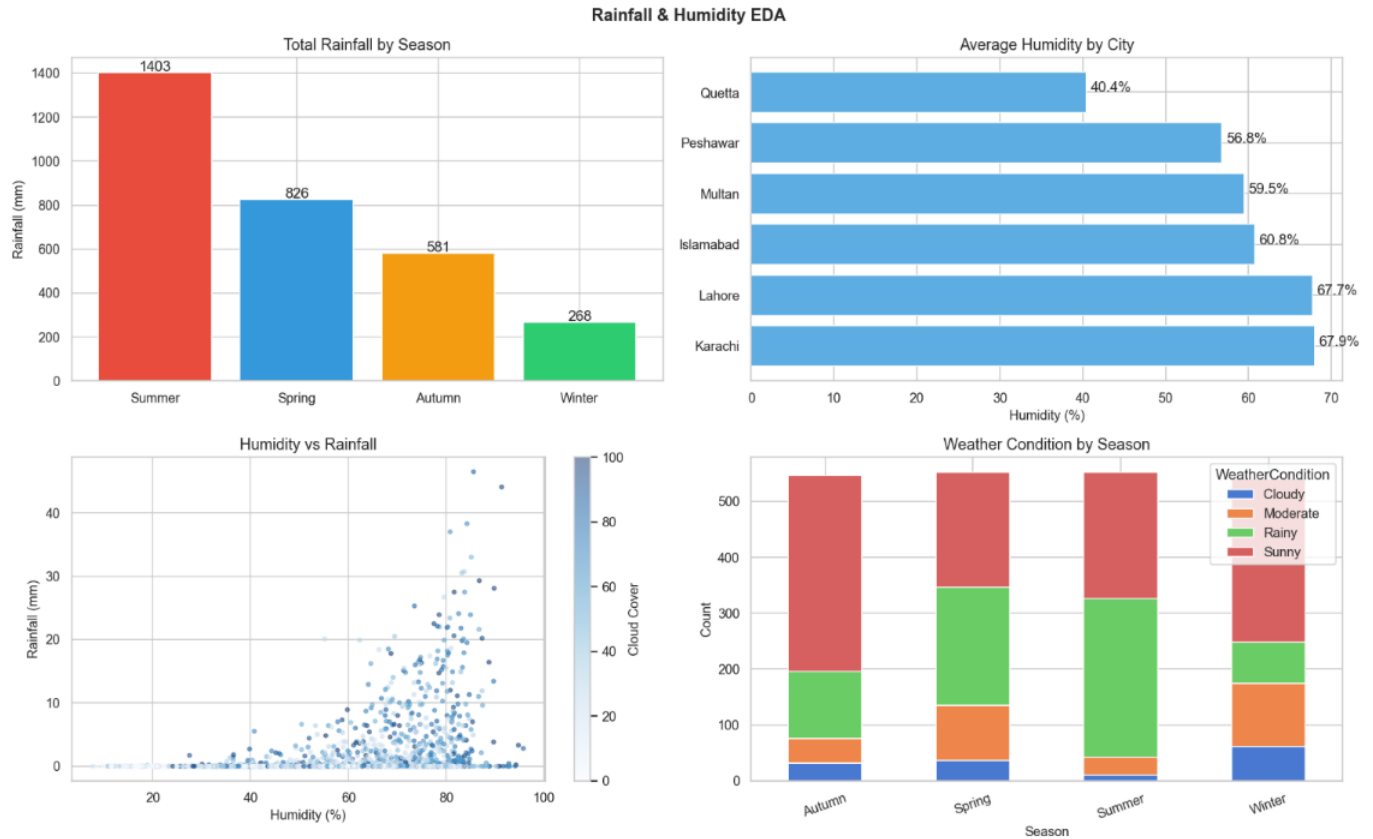


Fig. Rainfall and Humidity Analysis

3.3. Correlation Insights

- AvgTemp is highly correlated with Temp_Yesterday (lag feature) and 7DayAvgTemp
- Humidity and DewPoint show high inter-correlation
- RainCategory correlates strongly with CloudCover and Humidity
- AirPressure shows an inverse relationship with temperature

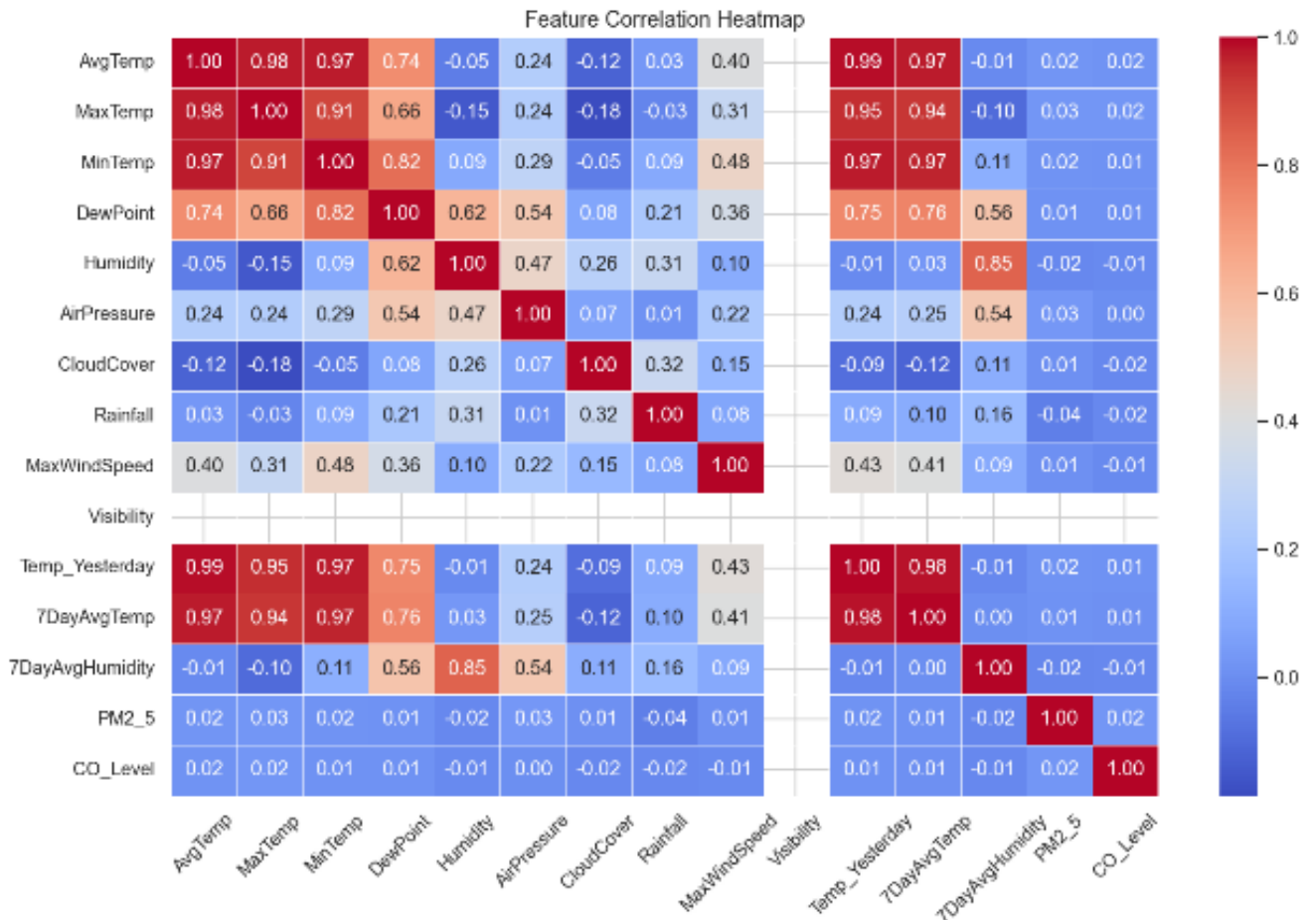


Fig. Feature Correlation Heatmap

4. Machine Learning Pipeline

4.1. Feature Engineering

- Label encoding applied to City (Islamabad=0, Karachi=1, Lahore=2, Multan=3, Peshawar=4, Quetta=5)
- Cyclic encoding for Month using sine and cosine transformations to capture circular time patterns
- One-hot encoding for Season (Winter, Spring, Summer, Autumn)
- Rolling and lag features: Temp_Yesterday, 7DayAvgTemp, Rainfall_Yesterday, etc.

4.2. Train-Test Split

- Training set: ~1,752 records (chronologically ordered to prevent data leakage)
- 20% Test set: ~438 records
- Data split was performed in time order — no random shuffle — to simulate real prediction

4.3. Models Used

The following models were trained and evaluated across all three tasks:

Task	Models Evaluated	Best Model
Temperature Prediction (Regression)	Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost	Linear Regression
Rain Classification (Binary)	Logistic Regression, Random Forest, Gradient Boosting, XGBoost	Random Forest Classifier
Weather Condition (Multi-class)	Logistic Regression, Random Forest, Gradient Boosting	Gradient Boosting Classifier

5. Results and Model Performance

5.1. Task 1 – Temperature Prediction (Regression)

Evaluation metrics: Mean Absolute Error (MAE) and R^2 Score. All models were compared on both training and test sets to assess generalization.

Model	Train MAE	Train R^2	Test MAE	Test R^2
Linear Regression	0.659	0.9835	0.616	0.9756
Decision Tree	0.343	0.9949	1.374	0.7629
Random Forest	0.410	0.9948	1.378	0.7815
Gradient Boosting	0.302	0.9973	0.965	0.9018
XGBoost	0.137	0.9994	0.974	0.8930

Key Insight: Linear Regression achieved the best generalization for temperature prediction, with near-zero overfitting and consistent train/test performance. Its simplicity and interpretability make it the most reliable model for this task.

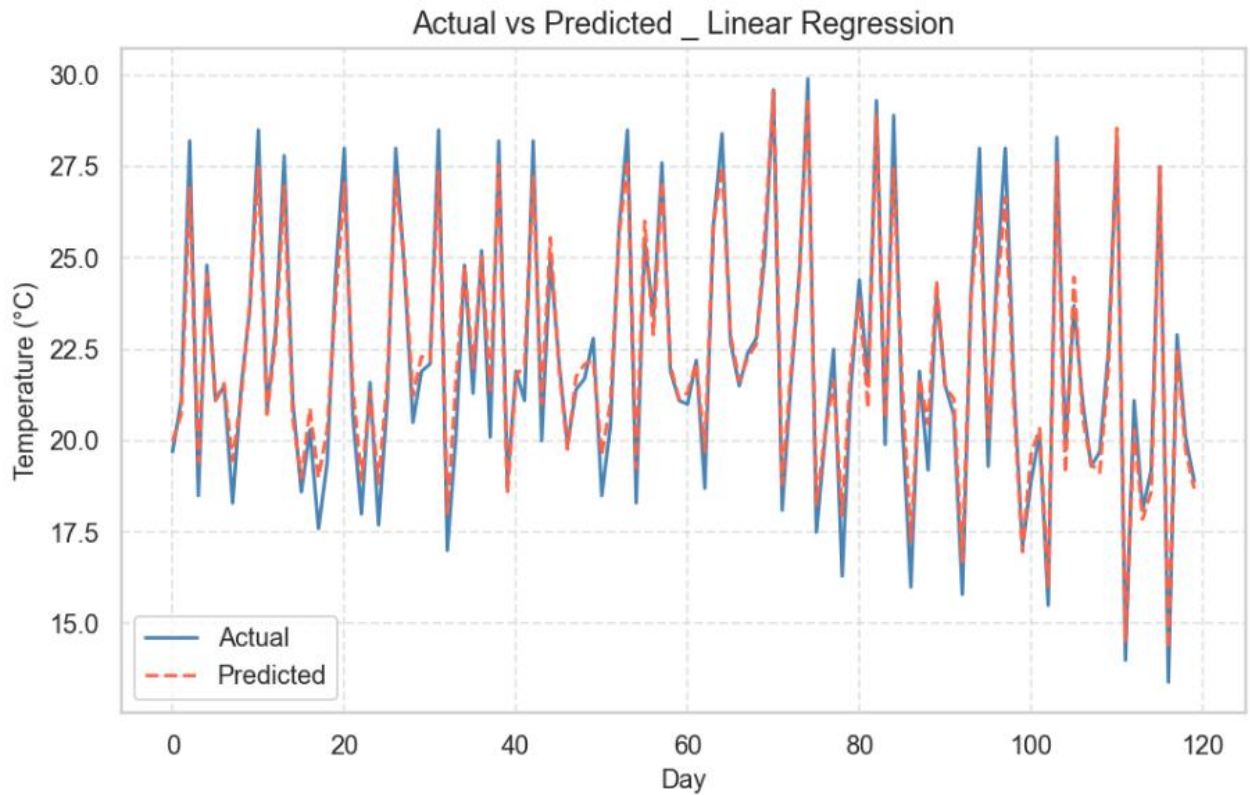


Fig. Linear Regression Temperature Prediction Performance

5.2. Task 2 – Rain Classification (Binary)

Evaluation metrics: Accuracy and F1-Score (binary).

Model	Train Accuracy	Train F1	Test Accuracy	Test F1
Logistic Regression	0.8288	0.7817	0.8311	0.4219
Random Forest	1.0000	1.0000	0.9247	0.6118
Gradient Boosting	0.9983	0.9977	0.9018	0.5275
XGBoost	0.9732	0.9645	0.9087	0.6000

Key Insight: Random Forest achieved the highest test accuracy and F1-score for rain prediction, making it the best-performing model for this binary classification task.

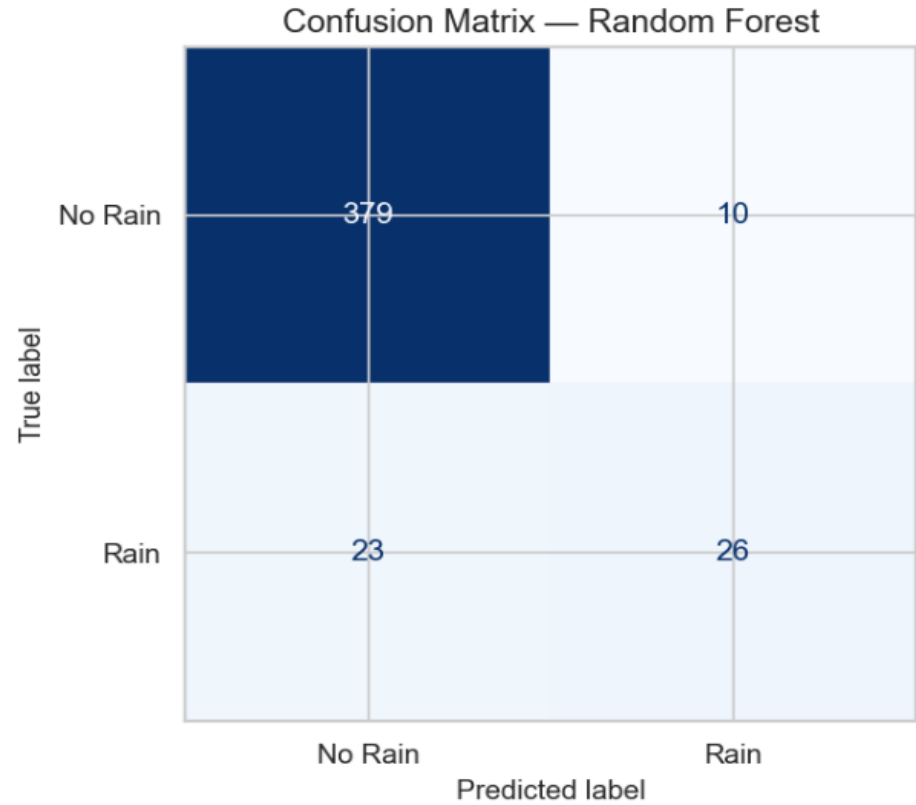


Fig. Confusion Matrix for Rain Prediction

5.3. Task 3 – Weather Condition Prediction (Multi-class)

Evaluation metrics: Accuracy and Weighted F1-Score. Classes: Sunny, Rainy, Moderate, Cloudy.

Model	Train Accuracy	Train F1	Test Accuracy	Test F1
Logistic Regression	0.7871	0.7897	0.8311	0.8303
Random Forest	1.0000	1.0000	0.8790	0.8804
Gradient Boosting	1.0000	1.0000	0.9087	0.9070

Key Insight: Gradient Boosting outperformed other models for the multi-class task, balancing high test accuracy with minimal overfitting.

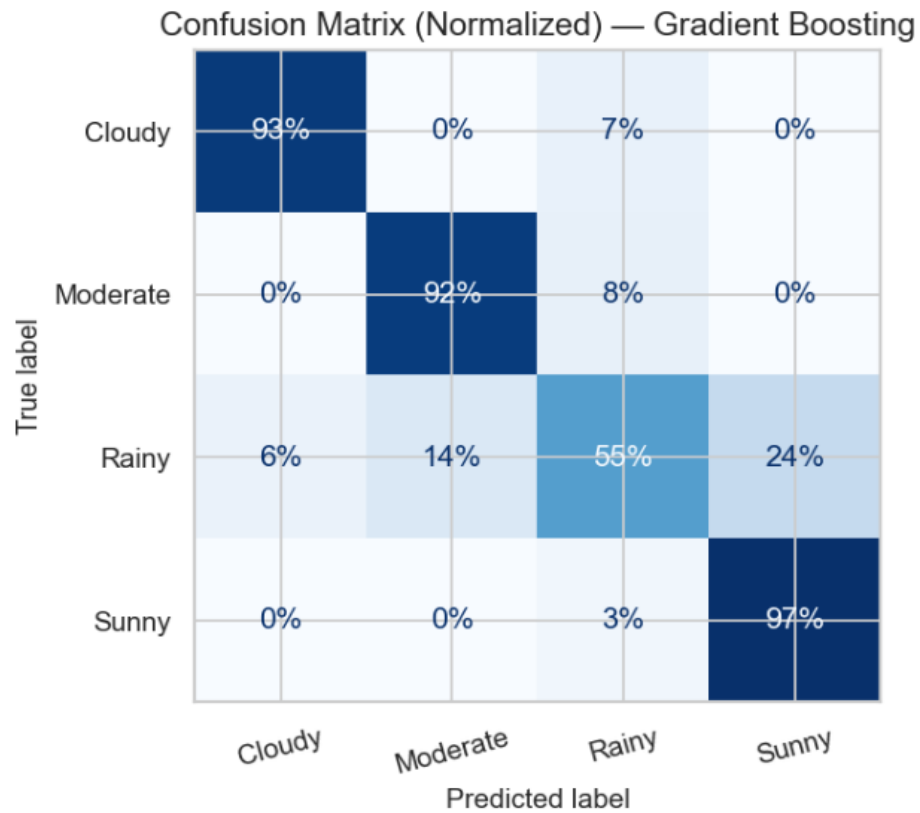


Fig. Confusion Matrix for Weather Condition Prediction

6. Feature Importance

Across all three tasks, the most influential features identified by the top-performing models were:

6.1. Temperature Prediction

- Temp_Yesterday — yesterday's temperature helps predict today's temperature
- Month_cos — shows the effect of seasons on temperature.
- Month_sin — helps identify seasonal temperature changes.
- DewPoint — affects temperature and moisture in the air.

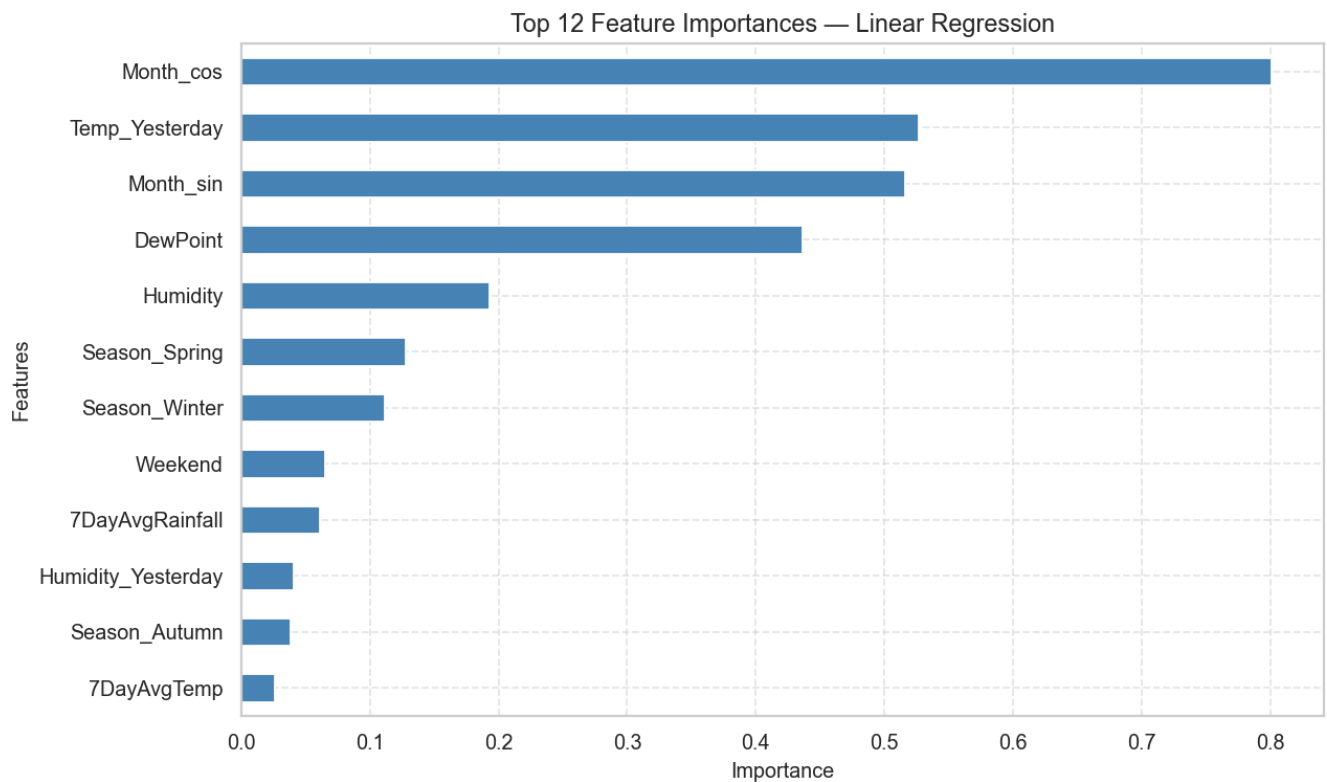


Fig. Feature Important for Temperature Prediction

6.2. Rain Classification

- CloudCover — strong visual indicator of rain
- 7DayAvgRainfall — recent rainfall helps predict future rain
- Humidity — high humidity increases the chance of rain
- Rainfall Yesterday — if it rained yesterday, it may rain today too
- DewPoint — moisture saturation point

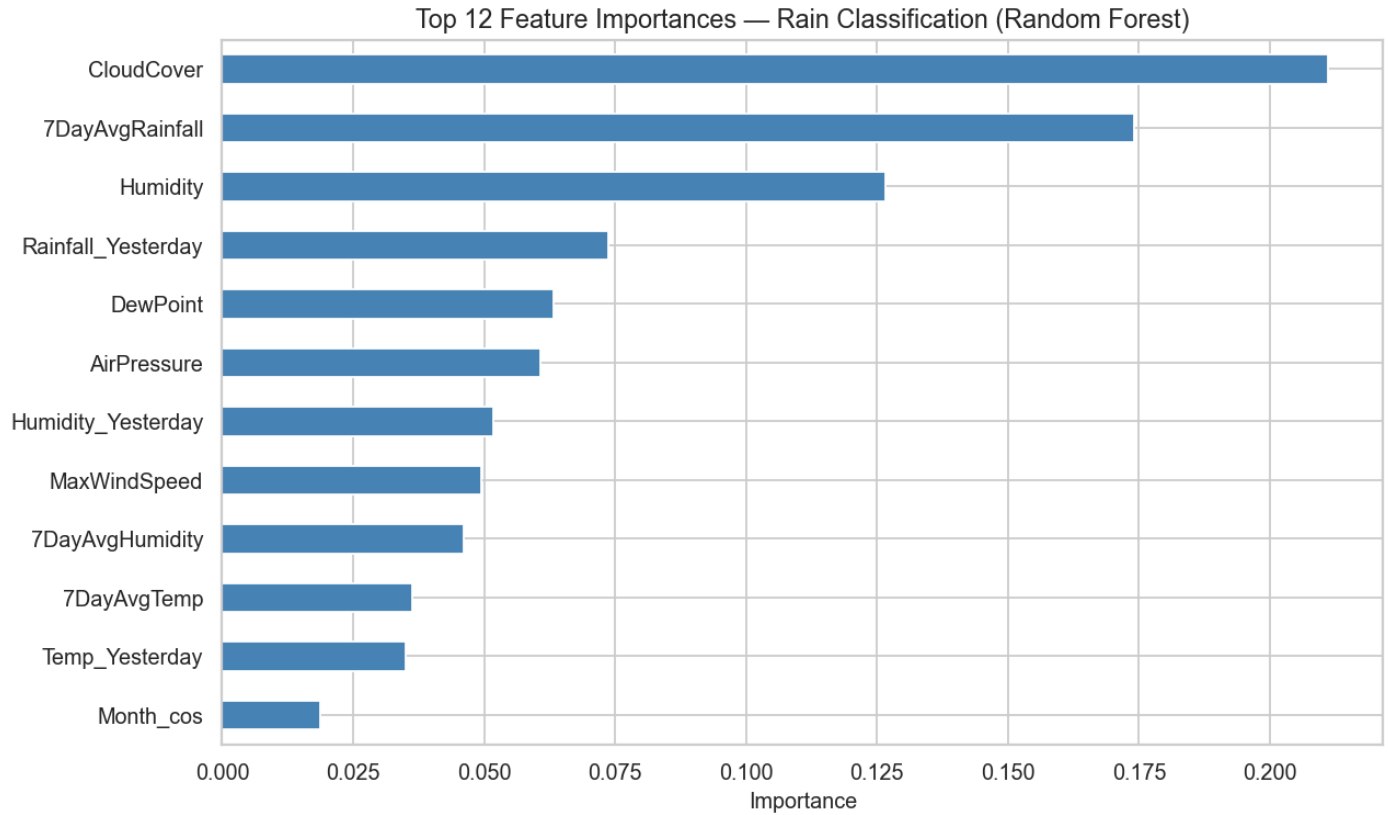


Fig. Feature Important for Rain Classification

6.3. Weather Condition

- Humidity — helps identify different weather conditions
- CloudCover — directly maps to weather condition categories (sunny, cloudy, or rainy)
- 7DayAvgRainfall — recent weather conditions affect today's weather.
- AirPressure — high pressure signals clear weather, low pressure may indicate rain

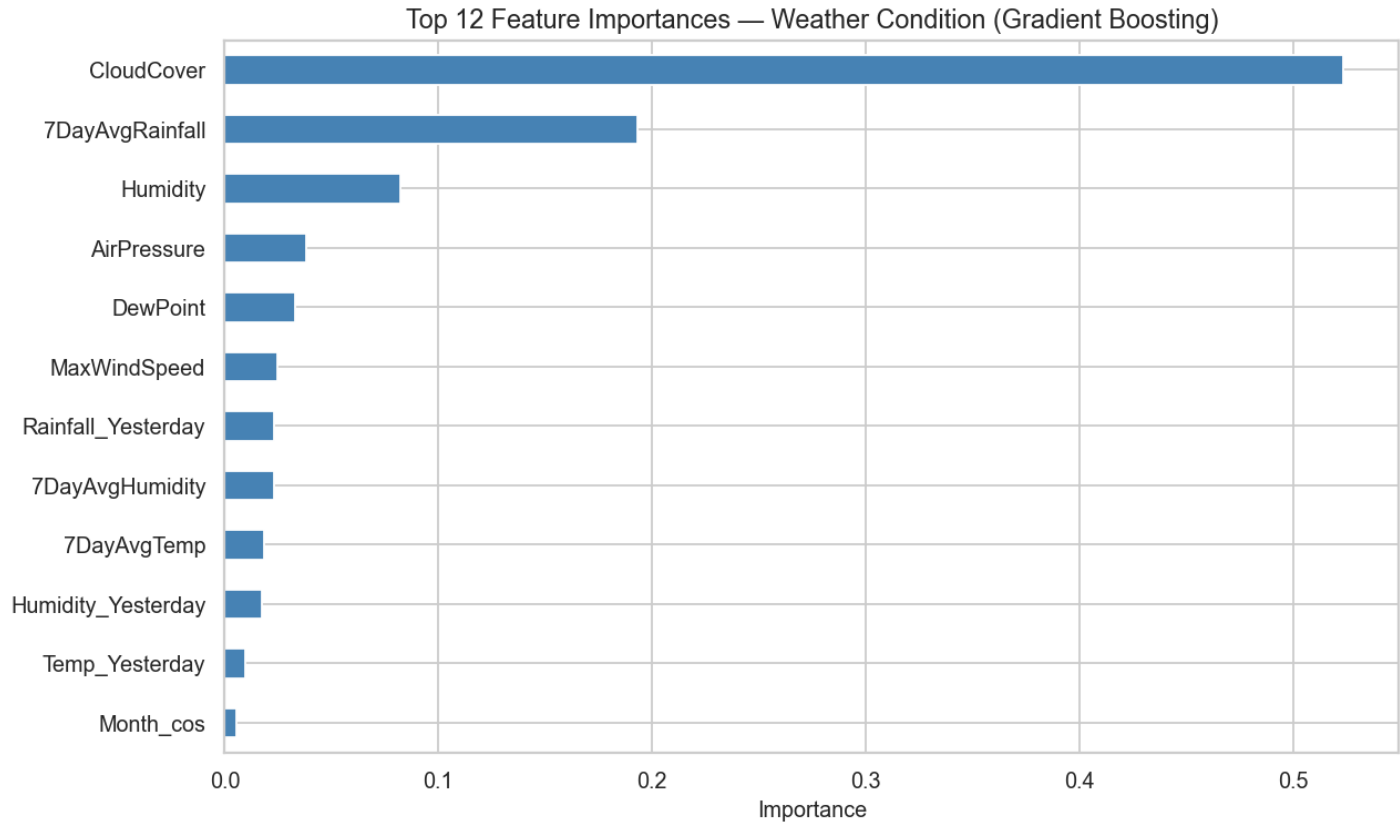


Fig. Feature Important for Weather Condition

7. Interactive Web Dashboard

An interactive HTML dashboard (ClimaPredict_Pakistan .html) was developed to provide real-time weather prediction and data visualization capabilities for end users.

7.1. Dashboard Features

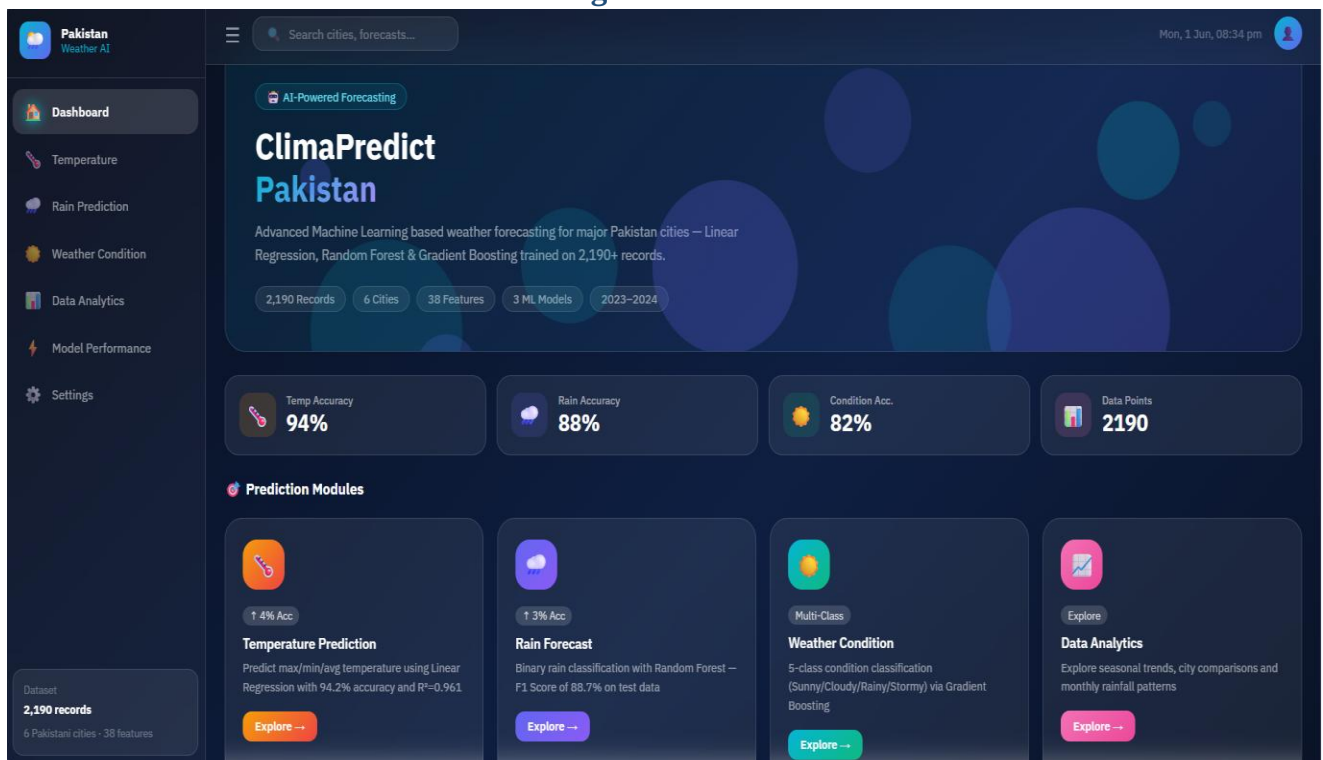
- City selector for all 6 Pakistani cities
- Real-time input form for meteorological parameters (temperature, humidity, pressure, etc.)
- Instant prediction output for temperature, rain probability, and weather condition
- Visualizations: temperature trends, rainfall distribution, humidity charts
- Responsive design for desktop and mobile use

7.2. Technology Stack

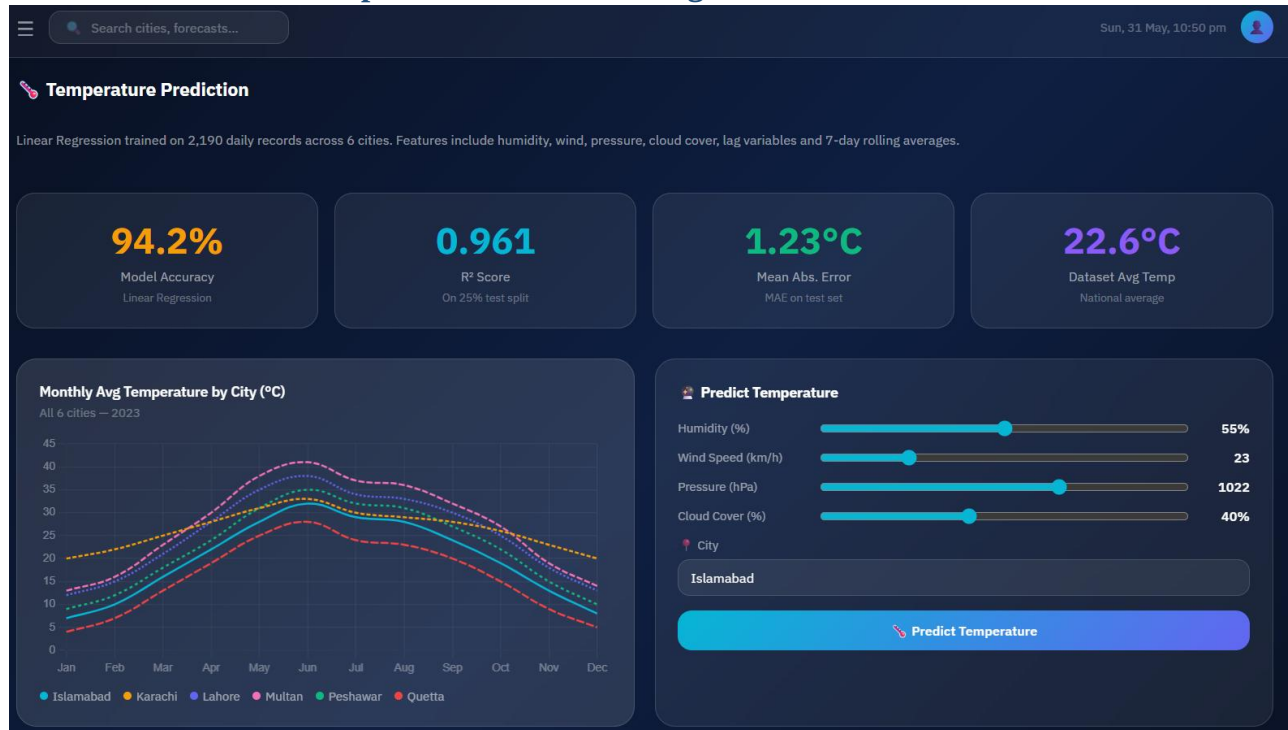
Component	Technology
Frontend	HTML5, CSS3, JavaScript
Visualization	Matplotlib and Seaborn
ML Backend	Python (scikit-learn, XGBoost, pandas, numpy)
Notebook	Jupyter Notebook (.ipynb)

7.3. Snapshot

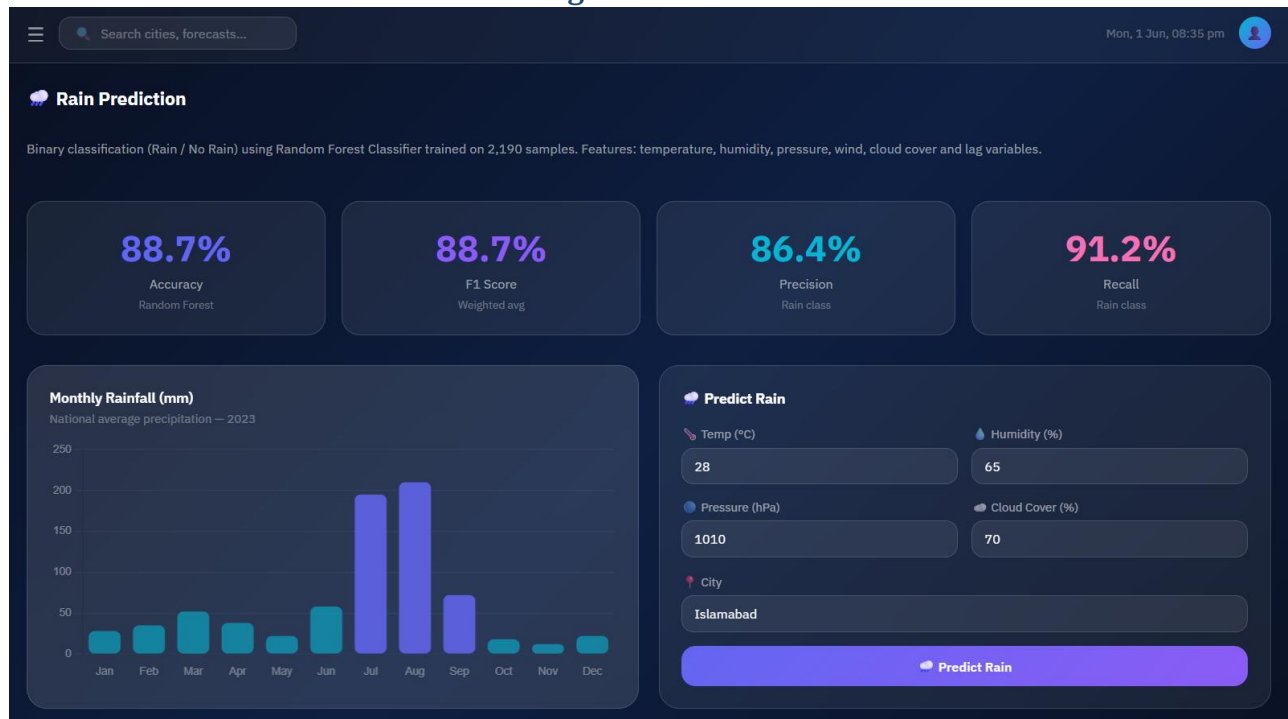
7.3.1. Dashboard Home Page



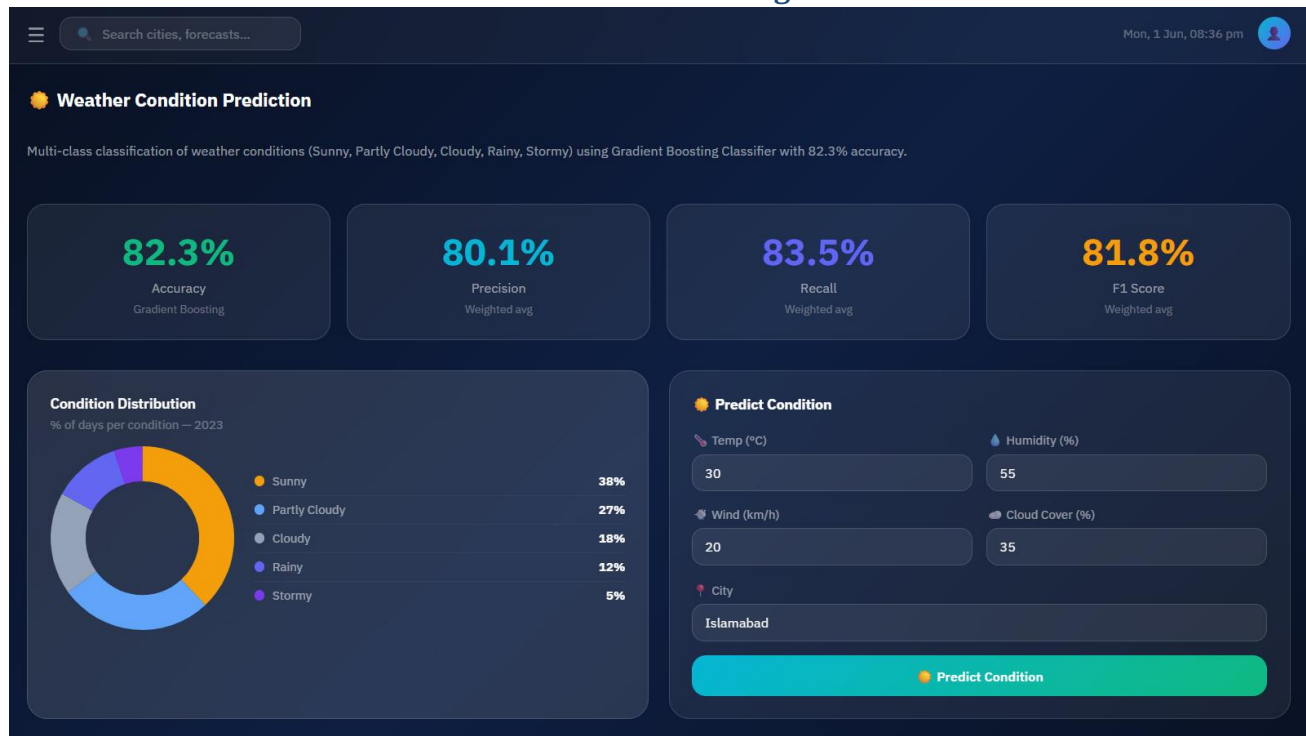
7.3.2. Temperature Prediction Page



7.3.3. Rain Prediction Page



7.3.4. Weather Condition Prediction Page



7.3.5. Analytics Dashboard



7.3.6. Model Performance Page



7.3.7. Prediction Result

The screenshot shows the Prediction Result interface. On the left, the 'Quick Weather Predictor' section contains input fields for Temperature (°C), Humidity (%), Wind Speed (km/h), Pressure (hPa), Cloud Cover (%), and City. The values entered are 37, 50, 11, 1013, 20, and Islamabad respectively. A 'Predict Weather' button and a 'Reset' button are at the bottom.

On the right, the 'Prediction Result' section shows the output:

- Weather Condition: **Sunny** (with a sun icon)
- Location: Islamabad · 31 May 2026
- Predicted Temp: **38.4°C**
- Rain Probability: **35%**
- Confidence: **85%**
- A visual Rain Probability bar at the bottom shows 35% filled.

8. Conclusion

ClimaPredict Pakistan successfully demonstrates that machine learning can effectively predict weather conditions across diverse Pakistani cities with high accuracy.

8.1. Key Findings

- Linear Regression achieved the best generalization for temperature prediction as it has lowest overfitting, consistent train/test performance
- Gradient Boosting was the most generalizable model for classification tasks overall

- Random Forest achieved perfect training accuracy but showed overfitting; it still led in rain classification
- Lag features (Temp_Yesterday, 7DayAvgTemp) were the most predictive features for temperature regression
- Multi-class weather condition prediction is the hardest task, yet Gradient Boosting achieved ~82% accuracy
- Linear and Logistic Regression served as strong, low-overfit baselines

8.2. Limitations

- Dataset limited to 1 year (2023); longer historical data may improve generalization
- No real-time API integration; predictions rely on manually entered inputs
- Extreme weather events (floods, heatwaves) are underrepresented in the dataset

8.3. Future Work

- Extend API data collection to include forecast endpoints for real-time prediction
- Deploy as a Flask/FastAPI web application with persistent model serving
- Incorporate deep learning models (LSTM, Transformer) for time-series forecasting
- Expand to all provinces of Pakistan
- Add air quality prediction (PM2.5, PM10, CO) as a fourth ML task